Median is the value which occupies the middle position when all the observations are arranged in an ascending or descending order. It is a positional average.

For example, the height of nine students in a class are 122 cm, 124 cm, 125 cm, 135 cm, 138 cm, 140cm, 141cm, 147 cm, and 161 cm.

(i) Usual calculation gives Arithmetic Mean to be 137 cm.

(ii) If the heights are neatly arranged in, say, ascending order, as follows 122 cm, 124 cm, 125 cm, 135 cm, 138 cm, 140cm, 141cm, 147 cm, 161 cm, one can observe the value 138 cm is such that equal number of items lie on either side of it. Such a value is called the Median of given readings.



(iii) Suppose a data set has 11 items arranged in order. Then the median is the 6th item because it will be the middlemost one. If it has 101 items, then 51st item will be the Median.

If we have an odd number of items, one can find the middle one easily. In general, if a data set has n items and n is odd, then the median will be the $\left(\frac{n+1}{2}\right)^{th}$ item.

(iv) If there are 6 observations in the data, how will you find the Median? It will be the average of the middle two terms. (Shall we denote it as 3.5^{th} term?) If there are 100 terms in the data, the Median will be 50.5^{th} term!

In general, if a data set has n items and n is even, then the Median will be the average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2}+1\right)^{th}$ items.

Example 8.8 The following are scores obtained by 11 players in a cricket match 7, 21, 45, 12, 56, 35, 25, 0, 58, 66, 29. Find the median score.

Solution

Let us arrange the values in ascending order.

0,7,12,21,25,29,35,45,56,58,66

The number of values = 11 which is odd

$$\text{Median} = \left(\frac{11+1}{2}\right)^{th} \text{ value}$$

$$= \left(\frac{12}{2}\right)^{th} \text{ value} = 6^{th} \text{ value} = 29$$

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